

ORIGINAL ARTICLE

Methods to address selection bias in post-trial studies of legacy effects were evaluated

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Abstract

Objectives: Post-trial follow-up studies have become increasingly important to investigate the long-term effectiveness of interventions after randomized controlled trials (RCTs). Legacy effects refer to intervention effects that are only observed after the trial has ended and are not the direct effects observed during the trial period. However, limited attention has been given to the potential selection bias in post-trial studies.

Methods: Using directed acyclic graphs, we illustrated potential sources of selection bias in post-trial studies of cardiovascular disease preventative interventions. We constructed scenarios where selection bias was present and undertook simulations to assess the ability of different modeling approaches to correct for this bias: no adjustment, adjustment for trial baseline covariates, adjustment for post-trial covariates and inverse probability weighting (IPW) methods. Using empirical data from the Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial and post-trial study, we demonstrated application of different modeling approaches.

Results: In the presence of selection bias, modeling without adjustment always resulted in biased estimates. Modeling with adjustment and IPW methods were able to correct the selection bias. The ACCORD study also demonstrated that while the direct effects were potentially beneficial, all models attempting to address selection bias revealed larger potential legacy effects when compared to unadjusted estimates.

Conclusion: Post-trial follow-up studies have the potential to provide valuable information for clinical practice by detecting legacy effects. However, it is important to consider and address selection bias that may arise from the post-trial study. This study highlights the importance of using an appropriate analysis method and identifying the potential bias sources to ensure that the findings are reliable and generalizable to the target population. © 2023 Elsevier Inc. All rights reserved.

Keywords: Legacy effects; Selection bias; Randomized controlled trials; Post-trial follow-up; Cardiovascular disease; Directed acyclic graphs

1. Background

Post-trial follow-up after randomized controlled trials (RCTs) allow for the evaluation of longer-term effects of an intervention. This type of study design has been used for a range of interventions including pharmaceutical products, surgery, screening for disease and health-promoting behaviors [1]. Unlike the direct effects of the intervention, which are evaluated in the original trial, “legacy effects” that may emerge subsequently are estimated in a post-trial analysis. The term is used to describe additional effects

that are observed when there is no longer a treatment contrast across randomized arms.

There has been particular interest in possible legacy effects of drugs to prevent cardiovascular disease (CVD), with a number of large-scale randomized controlled trials reporting the results of post-trial monitoring after the end of the initial trials [2–4]. For example, in a post-trial analysis on the United Kingdom Prospective Diabetes Study (UKPDS), a landmark RCT of glycaemic therapies in patients with newly diagnosed type-2 diabetes, researchers found increased benefit in myocardial infarction and all-cause mortality for the earlier improved glucose control in the intervention group, even though there was no longer any difference in antihyperglycaemic treatment or HbA1c levels across randomized arms [5,6]. Evidence of beneficial legacy effects such as this could strengthen the case for the early initiation of preventative treatment.

An analytic challenge of post-trial studies is that there is usually no longer a randomized comparison between the

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What is new?

- Selection bias in post-trial studies may result in biased estimates of legacy effects in unadjusted between-group comparisons.
- Through careful planning for prospective data collection in post-trial studies, valuable information on potential legacy effects can be obtained.
- Researchers aiming to investigate legacy effects should be aware of the potential for selection bias, and ensure they use an appropriate method of analysis.

intervention and control group, as analysis is done conditioning on the survivors enrolled in the post-trial study and who agree to participate in it. This type of study design is therefore a cohort study, which like other observational studies is at risk of selection biases. The selection bias in post-trial studies occurs when the participants selected for the post-trial study are not representative of the original randomized population. This may be due to differences across study groups in survival to end of the trial, participation in post-trial study, imbalances in covariates, and in medication use. These limitations have been infrequently mentioned in post-trial reports to date [7].

The objectives of this study are as follows: first, to use directed acyclic graphs (DAGs) to illustrate the concept of legacy effects and the potential for selection bias in post-trial follow-up studies after RCTs to prevent cardiovascular disease [8]. The directed acyclic graph (DAG) is a visual representation of complex causal relationships between variables, which can help researchers design studies and identify potential biases. Second, to conduct simulations to quantify the extent of bias caused by varying treatment effects, underlying absolute CVD risk, and causal pathways for legacy effect, and to compare the performances of different modeling strategies. Third, to investigate these modeling strategies using the real-world data from post-trial follow up of Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial.

2. Methods

2.1. Notation and definition

Consider an intervention A and outcome Y , and a measured covariate L that represents the severity of disease (summary of risk). We assume participants enter a randomized controlled trial at time zero (t_0) until the trial ends (t_1) unless they have an event of interest. After the trial, the surviving participants who have not had the event of interest are followed until the post-trial study ends (t_2). Therefore,

L_0 represents the measured baseline risk, and A_0 represents the intervention or comparator assigned at the start of the trial; due to randomization, baseline risk and intervention type are independent of each other. L_1 represents the measured risk after the trial (start of post-trial period) and A_1 is the post-trial choice of treatment. Y_1 and Y_2 denote the outcomes at the end of the initial trial and at the end of post-trial follow-up, respectively.

Legacy effects generally refer to the effects of an intervention on the health outcomes observed after the end of the trial (i.e., $A_0 \rightarrow Y_2$ causal effect), that are not due to the direct effects observed within the trial period ($A_0 \rightarrow Y_1 \rightarrow Y_2$ causal pathway) (Fig. 1). As illustrated, there are at least three different causal pathways through which a legacy effect could occur. One possible pathway is where the trial intervention has an impact on the choice of intervention during post-trial period ($A_0 \rightarrow A_1$). For example, in some health promotion programs (e.g., smoking cessation, low-salt diet), participants randomized to the intervention may be more likely to have healthy behaviors after the end of the trial. A second possible pathway is where the trial intervention influences the measured post-trial risk, even where there is no direct impact on the choice of post-trial intervention ($A_0 \rightarrow L_1$). A third possible pathway is where the intervention influences long term outcomes ($A_0 \rightarrow Y_2$). This last causal pathway is the most common type of legacy effect examined in post-trial studies of drugs to prevent cardiovascular disease. Henceforth, we refer only to this third pathway as the legacy effect.

2.2. DAGs of different modeling strategies

We assumed a setting of a post-trial study after an RCT of a CVD prevention drug treatment and a comparator. Participants return to their usual system of health care, with no attempt to maintain previously randomized therapies. The medication use is according to their clinical needs, basing on the findings of the trial or existing guidelines (we assumed no pathway from $A_0 \rightarrow A_1$). We created two scenarios where (i) trial treatment had no effect on participants' post-trial absolute cardiovascular risk and (ii) treatment lowered participants' post-trial absolute cardiovascular risk (i.e., absence or presence of $A_0 \rightarrow L_1$ causal pathway). Selection bias may result from conditioning on the surviving participants without clinical CVD, when the trial treatment has a direct effect on this outcome. For example, this may occur where intervention group were less likely to have a CVD event than the comparator group. The CVD outcome (Y_1), is a collider which opens up back door paths (through L_0 and L_1) [9]. Here we build upon this common structure for selection bias and demonstrate how it could affect the post-trial analysis under different modeling strategies.

Figure 2 shows the DAGs for different modeling strategies of the two scenarios. In Diagram a), no adjustment is made to correct the selection bias. The presence and degree

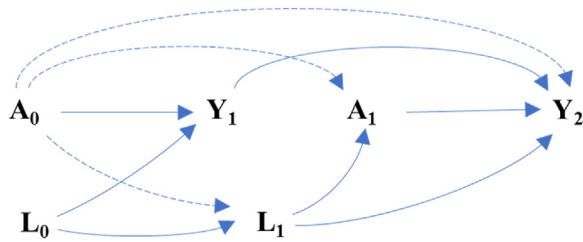


Fig. 1. An illustration of legacy effect through directed acyclic graph. The dashed lines are potential pathways for a legacy effect. L_0 represents the baseline risk, and A_0 represents the intervention assigned at the start of the trial. L_1 represents the risk after the trial (start of post-trial period) and A_1 is the post-trial choice of treatment. Y_1 and Y_2 denote the cardiovascular outcomes at the end of the initial trial and at the end of post-trial follow-up, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

of the selection bias depends on if the treatment has a direct effect on the trial outcome and the size of the effect. In Diagram b), if the measured post-trial cardiovascular risk is influenced by measured baseline risk ($L_0 \rightarrow L_1$), but independent of treatment received during the trial, then the selection bias induced by conditioning on the trial outcome might be controlled by adjusting for baseline risk L_0 . However, if trial treatment A_0 also influences measured post-trial cardiovascular risk L_1 , then the estimate remains biased due to the pathway $A_0 \rightarrow L_1 \rightarrow Y_2$. That is, the adjustment for measured baseline cardiovascular risk L_0 will not account for the effect on the long term outcome (post-trial clinical cardiovascular disease) that is mediated

through the intermediate variable L_1 (measured post-trial cardiovascular risk). In Diagram c), the potential selection bias has been adequately accounted for by adjusting on the post-trial cardiovascular risk L_1 . An expansion of the DAGs to include unmeasured confounders, and implications for the modeling strategies, is included in the [Supplementary Materials](#).

In addition to these conventional modeling strategies, the inverse probability weighting approach has been used to handle selection bias due to censoring in observational studies. This approach creates a “pseudo-population” in which nobody is censored by assigning inverse probability weights—the censored individuals are replaced by copies of the uncensored individuals with similar characteristics, thus eliminating the bias [10].

2.3. Simulations to compare different modeling strategies

We conducted simulations to compare different modeling strategies and explore influencing factors on the selection bias. The setting of simulation combined a randomized controlled trial and a post-trial follow-up study, to broadly reflect data that may be observed in a clinical trial for cardiovascular disease prevention. We generated independent data for different scenarios and then evaluated the four modeling strategies by applying each of them to the simulated data. The method and R code for the data generation are provided in the [Supplementary Materials](#).

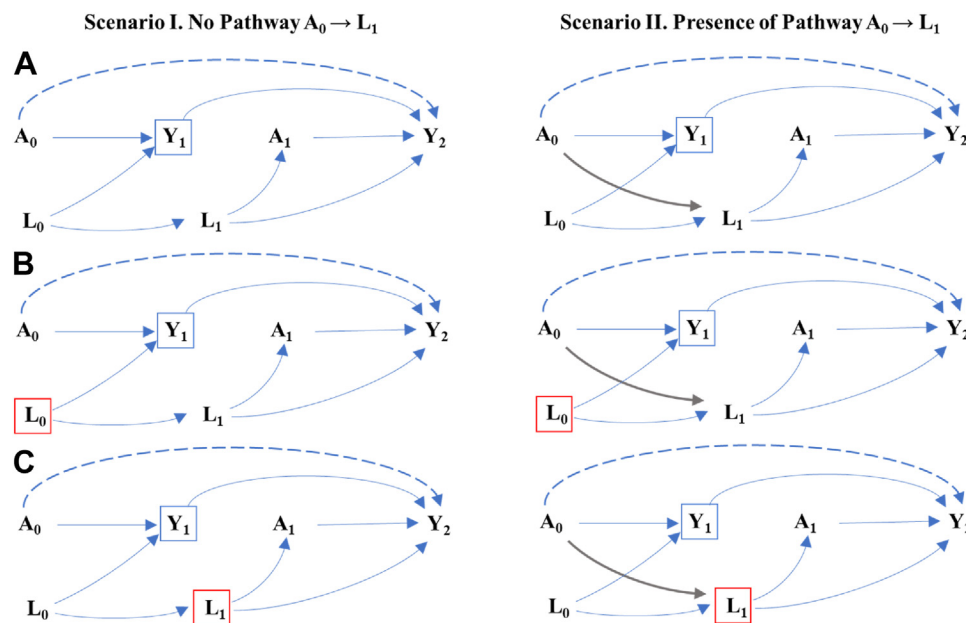


Fig. 2. Causal diagrams for different modeling strategy. The dashed lines are the legacy effects on interest. L_0 represents the baseline risk, and A_0 represents the intervention assigned at the start of the trial. L_1 represents the risk after the trial (start of post-trial period) and A_1 is the post-trial choice of treatment. Y_1 and Y_2 denote the cardiovascular outcomes at the end of the initial trial and at the end of post-trial follow-up, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Simulation settings were divided into two main scenarios, according to whether or not the trial treatment affects post-trial cardiovascular risk (i.e., presence or absence of $A_0 \rightarrow L_1$ causal pathway). In the first scenario, we assumed that treatment did not affect measured post-trial cardiovascular risk (the pathway $A_0 \rightarrow L_1$ does not exist) and that post-trial risk (L_1) was only influenced by baseline risk (L_0). In the second scenario, we assumed that treatment lowered post-trial cardiovascular risk (the pathway $A_0 \rightarrow L_1$ exists), and the patients who received treatment during the trial would have 10% reduction of the post-trial risk (post-trial risk was still also influenced by baseline risk). For simplicity, we assumed there are no unmeasured confounding, measurement error, nonadherence, or loss to follow-up.

Table 1 provides a summary of the variables considered in the simulations. We varied the level of absolute risk (i.e., the 5-year survival rate), the size of the direct treatment effect and the size of the legacy effect across simulations. The parameters were chosen so that approximately 90%, 80%, and 70% of participants who receive no treatment will survive without clinical CVD after the 5-year RCT. For greater generalizability, the distribution of baseline risk L_0 for participant was simulated with a log-normal distribution. The treatment effects (hazard ratios) were defined as 0.9, 0.7, and 0.5, representing a 10%, 30%, and 50% lower risk of clinical CVD compared with the control group. The size of the legacy effect was defined as 1 (no legacy effect), 0.9, and 0.8. This resulted in a total of 54 scenarios, and 1,000 simulations were run for each scenario.

We compared four different approaches to modeling: i) without any adjustment; ii) with adjustment for baseline cardiovascular risk L_0 ; iii) with adjustment for post-trial cardiovascular risk L_1 if available, and iv) using inverse probability weighting (IPW). In the IPW method, the weights were calculated by the inverse of probability of survival by the end of the trial, and the survival probability was estimated by the baseline cardiovascular risk L_0 and treatment within the trial A_0 . These weights were then applied to the observed post-trial population to account for the selection bias by creating a new pseudo-population. Depending on whether the measured post-trial cardiovascular risk of participants was available or not, the analysis of the pseudo-population was conducted in an adjusted and unadjusted manner. Participants were

analyzed in the groups to which they were randomly allocated at the start of the initial clinical trial, and Cox proportional hazard models were fitted using these four approaches.

3. Results

The estimated legacy effect (mean of hazard ratios) using the different approaches are presented in [Figures 3 and 4](#). In the scenarios where trial treatment did not affect measured post-trial cardiovascular risk (the pathway $A_0 \rightarrow L_1$ does not exist), the estimates without making any adjustment were most biased and the degree of the bias increased with increasing size of the direct drug effect and underlying absolute risk of the study population. The estimates with adjustment for baseline risk were similar to those with adjustment for post-trial risk and inverse probability weighting (post-trial data available) in most simulation settings. The inverse probability weighting method using only the baseline data also reduced the selection bias compared to no adjustment, and performed especially well in the scenarios with lower underlying absolute risk.

In the scenarios where trial treatment did affect measured post-trial cardiovascular risk (the presence of pathway $A_0 \rightarrow L_1$), both the adjusted method and inverse probability weighting method using the post-trial risk had good performance on the estimation for legacy effects. In contrast, the other methods (no adjustment, baseline risk adjusted, IPW where post-trial data not available) tended to overestimate the size of the legacy effect.

3.1. Example: ACCORD and its follow-up study

The Action to Control Cardiovascular Risk in Diabetes (ACCORD) study was a two-by-two factorial randomized controlled trial, which evaluated the effects of intensive glycemic control, intensive blood pressure control, and fibrate treatment, on the prevention of cardiovascular disease in people with Type Two Diabetes Mellitus (T2DM). It enrolled 10,251 people (mean age 62 years), who had a history of T2DM for a median duration of 10 years. Participants had either a history of previous cardiovascular disease or had elevated risk factors levels. After a median follow-up of 3.7 years, the glycemic trial was stopped early because intensive glucose control was associated with increased all-cause mortality [11]. The Blood Pressure Trial and Lipid Therapy Trial were completed as planned, and are the focus of our illustrative example using the modeling strategies outlined above. In the Blood Pressure Trial, BP lowering drugs targeting a systolic blood pressure of less than 120 mm Hg, as compared with less than 140 mm Hg, did not reduce the rate of a composite outcome of fatal and nonfatal major cardiovascular events [12]. In the Lipid Therapy Trial, the combination of fenofibrate and simvastatin did not reduce the rate of fatal cardiovascular events,

Table 1. Summary of variables used in the simulation

Variable	Value
Sample size	8,000
Length of RCT and PTFU	5 yr, 5 yr
5-yr survival rate	0.7, 0.8, 0.9
Individual risk distribution	log-normal (mean = 0, SD = 0.5)
Initial Treatment effect	0.9, 0.7, 0.5
Size of legacy effect	1 (no legacy effect), 0.9, 0.8

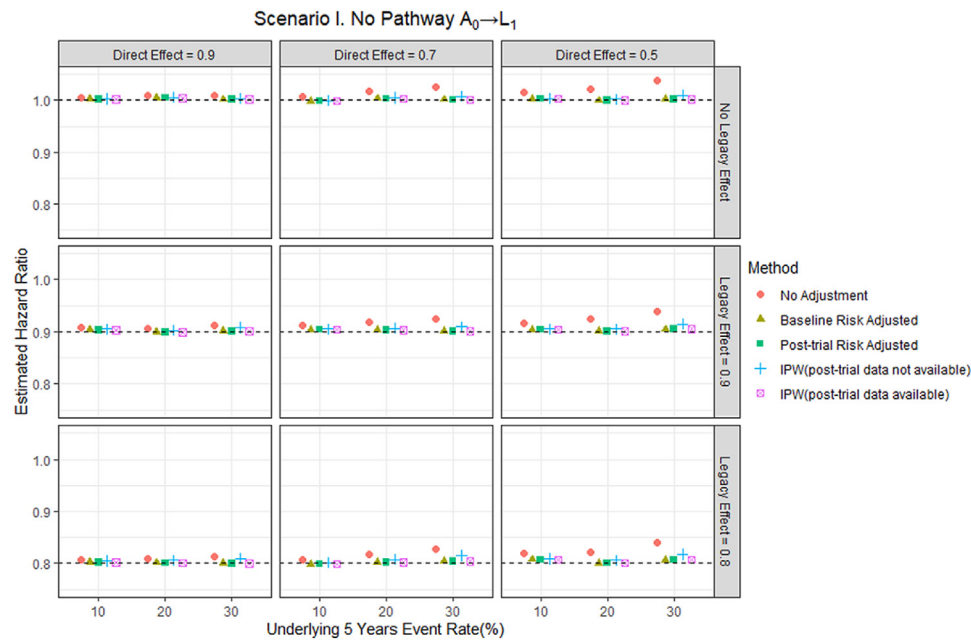


Fig. 3. Estimated legacy effects (hazard ratios) using the different modeling strategies when the pathway $A_0 \rightarrow L_1$ does not exist. Dashed lines are the true legacy effects in each scenario. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

nonfatal myocardial infarction or nonfatal stroke, as compared with simvastatin alone [13]. However, a possible benefit was found for patients with dyslipidemia (both a high baseline triglyceride level and a low baseline level of high-density lipoprotein cholesterol). The direct treatment effect on total mortality in these three study

populations (BP Trial, Lipid Trial—all, Lipid Trial—dyslipidemia subgroup) and the percentage of enrollment into the post-trial period are shown in Table S1. The blood pressure and lipid post-trial studies enrolled similar proportion of participants across the two original groups, while in the dyslipidemia subgroup post-trial study a higher

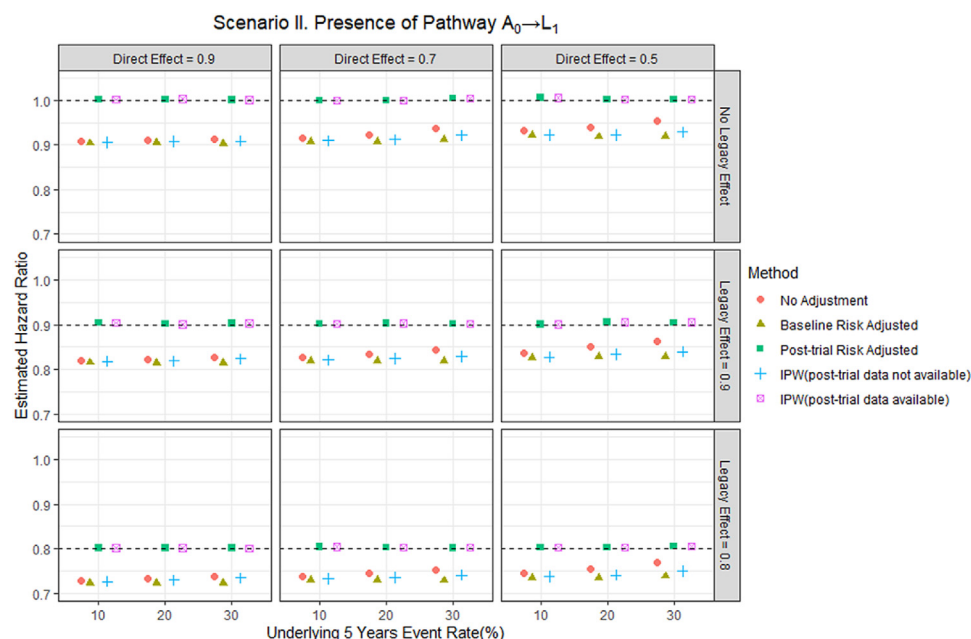


Fig. 4. Estimated legacy effects (hazard ratios) using the different modeling strategies when the pathway $A_0 \rightarrow L_1$ exists. Dashed lines are the true legacy effects in each scenario. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2. Estimated legacy effects using the different modeling strategies (hazard ratios and 95% confidential intervals)^a

Study	Modeling strategies of post-trial analysis				
	Unadjusted	Baseline covariate adjustment	Post-trial covariate adjustment	IPW	
				Post-trial covariate not available	Post-trial covariate available
BP Trial	1.03 (0.86, 1.24)	0.99 (0.82, 1.20)	0.97 (0.62, 1.50)	1.00 (0.83, 1.22)	0.99 (0.62, 1.58)
Lipid Trial	0.95 (0.81, 1.11)	0.94 (0.80, 1.10)	0.87 (0.64, 1.17)	0.96 (0.82, 1.13)	0.92 (0.72, 1.18)
Lipid Trial Dyslipidemia Subgroup	0.64 (0.45, 0.91)	0.61 (0.42, 0.89)	0.54 (0.30, 0.99)	0.57 (0.35, 0.94)	0.56 (0.29, 1.12)

^a Estimated legacy effects of treatments on total mortality in ACCORD. The covariates that were adjusted for included age, gender, race, education, CVD history, clinical center, smoking status duration of diabetes, HbA1c, systolic blood pressure, total cholesterol, and HDL-C. The probability of survival in the IPW method was estimated using the above covariates and trial treatment assignment.

percentage of patients from active group were enrolled than the placebo group. The comparisons of other variables were provided in the [Supplementary Materials](#).

We compare the different modeling strategies by applying them to these three sets of trial data. [Table 2](#) shows the estimated legacy effects of treatments on total mortality in ACCORD. The covariates that were adjusted for were: age, gender, race, education, CVD history, clinical center, smoking status duration of diabetes, HbA1c, systolic blood pressure, total cholesterol and HDL-C. The probability of survival in the inverse probability weighting method was estimated using the above covariates and trial treatment assignment.

In the BP and Lipid trial-all, where the direct effects were found to be neutral, the unadjusted estimates for legacy effects were similar to those of other methods. In the dyslipidemia subgroup, where the direct effects were found to be possibly beneficial, all the models aiming to correct the selection bias show larger potential legacy effects compared with the unadjusted estimates. These findings are consistent with the simulation results in [Figure 3](#), when the pathway $A_0 \rightarrow L_1$ does not exist (i.e., combination of fenofibrate and simvastatin during the trial did not affect measured post-trial cardiovascular risk). This suggests the potential legacy effects of the intervention on long term outcomes ($A_0 \rightarrow Y_2$), if present, occurs through other causal pathways.

4. Discussion

In this paper, we used directed acyclic graphs (DAGs) to illustrate the concept of legacy effects and the causal pathways through which legacy effects may operate. Although the setting of the simulation and example used were focused on the CVD drug treatment, the methodological considerations may also apply to other interventions.

We have highlighted the potential selection bias due to conditioning on the survivors enrolled in the post-trial period. However, there may be minimal selection bias if the trial treatment does not affect the measured post-trial risk, or the event of interest is less common (lower

underlying risk in the study population), as the treatment arms could be expected to remain almost balanced for covariates (Scenario I). We previously found that the median proportion of trial participants who were enrolled in post-trial studies of a CVD preventive intervention was about 90%, and that there was balance of covariates across study groups where this was measured [7]. When post-trial data on covariates are not available or not well-collected, adjusting for the baseline covariates may be used to correct the selection bias. We found that it may have a similar performance to adjusting for post-trial covariates when the trial assignment does not affect the post-trial risk/covariates.

In the second scenario, when we assume treatment influences the post-trial risk, the methods not using post-trial covariate data gave biased results. Given sufficient post-trial data, adjustment of the post-trial covariates could provide unbiased estimates, which removes the potential influence of the intermediate variable. When post-trial data is not available, it is difficult to identify the causal pathway for an apparent legacy effect. It may occur where the post-trial risks have been modified through the intermediate variable, or the trial assignment has an impact on the choice of treatment during post-trial period, or there exist other unknown causal pathways. The post-trial data could provide information to infer the mechanism of legacy effect, and investigators should be aware of the limitation of the estimation for legacy without post-trial data. We also recommend performing sensitivity analyses to assess how different assumptions influence the effect estimates.

We found the inverse probability weighting approach could also be used to correct the selection bias. When the post-trial data were not available, it did not show superior performance compared with the traditional adjustment methods in the first scenario [14,15]. Where post-trial data were available, the inverse probability weighting method showed similar performance with the post-trial adjusted method. Inverse probability weighting and other ‘G-methods’ for causal inference have been used in observational studies to handle analytic challenges such as time-variant covariates and treatment-confounder feedback [16]. However, the most common methods currently used in post-trial studies are same as the conventional statistical

models used in the primary trials, and more flexible models have rarely been used [7]. We anticipate that the causal inference methods may have better performance in the analysis of real-world data with more complex data [17,18].

Our study has some limitations. We did not consider unmeasured confounders, such as genetic or environmental risk factors, which may influence the results of post-trial analysis. The robustness of the results with respect to unmeasured confounding should be assessed using sensitivity analyses [19]. For simplicity, we did not consider treatment switching and lost to follow-up in the simulation, and the scenarios we investigated cannot cover all situations that may occur in reality. Careful thought incorporating clinical expert knowledge on the causal pathways is necessary when applying these methods to real datasets.

5. Conclusion

In conclusion, the choice of modeling strategy of post-trial analysis should consider the availability of data, type of intervention, causal pathway of legacy effect, and size of direct effect. We also recommend performing sensitivity analyses to assess how different assumptions and unmeasured confounders may influence the effect estimates. More flexible models may be used to address complex issues that arise in real world post-trial studies.

CRedit authorship contribution statement

Lin Zhu: Conceptualization, Investigation, Methodology, Writing — original draft. **Katy J.L. Bell:** Conceptualization, Investigation, Methodology, Supervision, Writing — review & editing. **Andrew Hayen:** Conceptualization, Investigation, Methodology, Supervision, Writing — review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2023.05.028>.

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